

Probabilistic Revenue Forecasting for E-commerce Marketing

An AI-Assisted Forecasting Utility for Digital Marketing Agencies

Introduction

E-commerce businesses invest across multiple acquisition channels such as Google Ads, Meta Ads, Microsoft Ads, organic search, affiliate networks, and display campaigns. Digital marketing agencies managing these businesses are expected to estimate future business outcomes before budgets are deployed. This is a difficult forecasting problem.

Marketing performance is influenced by seasonality, changing user behavior, inconsistent campaign structures, varying channel efficiency, and incomplete analytics visibility. Existing forecasting workflows are often spreadsheet-driven, manually maintained, and disconnected across platforms.

This hackathon challenges participants to build a practical AI-assisted forecasting utility that predicts ecommerce revenue and ROAS using historical analytics and sales data.

The solution should focus on realistic business forecasting rather than theoretical modeling, while clearly communicating assumptions, uncertainty, and operational insights.

Why This Solution Needs to Be Built

Most ecommerce forecasting today suffers from three major problems:

1. Many forecasting systems optimize for traffic or conversions without considering ROAS constraints.
2. Marketing data is fragmented and inconsistent. Agencies work with multiple paid channels, manually tagged campaigns, and varying reporting structures across platforms.
3. Forecasts rarely explain why performance is expected to improve or decline, making strategic planning difficult.

At the same time, agencies increasingly need forward-looking decision support instead of retrospective reporting.

Challenge Description

Participants are required to build a working prototype that forecasts future ecommerce performance using historical marketing and sales data.

The solution should generate probabilistic forecasts for:

- expected ecommerce revenue (aggregate of Google Ads, MS Ads, and Meta Ads revenue)
- expected blended ROAS,

- channel-level / campaign-type / campaign-level revenue contribution,
- channel-level / campaign-type / campaign-level ROAS ranges.

The forecasting window should support aggregate planning periods between 30, 60 and 90 days.

The system should use:

- Google Ads, MS Ads, and Meta Ads conversion data,
- For campaign type forecasting, consider seasonality based on available data
- Use additional functions to forecast revenue based on different media budgets.

The solution should support forecasting across channels.

Participants are expected to use existing channel-level attribution data as-is. Building a custom attribution engine or a full-scale Media Mix Modeling (MMM) is outside the scope of this challenge.

The forecasting system should also incorporate an AI-assisted causal inference layer using LLM APIs (such as OpenAI, Gemini, Anthropic or similar); LLMs may also be used for anomaly interpretation, forecast explanation, and identifying operational risks.

The focus of the challenge is not building a full SaaS platform, but creating a technically sound and operationally realistic forecasting utility.

Constraints

Participants should work within the following constraints:

- Forecasts should be aggregate-period forecasts rather than daily forecasts.
- Forecasts should be probabilistic ranges instead of single deterministic values.
- Revenue and ROAS are the primary business metrics to forecast.
- Existing attribution should be treated as the source of truth.

The objective is to build a realistic and explainable forecasting prototype within the hackathon timeline.

Deliverables

Participants are required to submit the following:

1. Working Prototype

A functional prototype capable of:

- ingesting channel level datasets (provided as .csv file),
- validating campaign consistency,
- accepting future media budget inputs,
- generating probabilistic revenue and ROAS forecasts,
- producing channel-level / campaign-type / campaign-level forecast outputs,
- generating AI-assisted causal summaries.

2. Technical Documentation

A concise document explaining:

- forecasting methodology,
- model selection,
- data preprocessing logic,
- assumptions,
- limitations,
- AI integration strategy.

3. Architecture Overview

A brief overview of:

- frontend stack,
- backend stack,
- forecasting pipeline,
- LLM integration workflow.

4. Demo Workflow

A short walkthrough or presentation showing:

- data ingestion,
- forecast generation,
- budget simulation,
- AI-generated business insights.

Evaluation Criteria

Submissions will be evaluated based on the following criteria:

Technical Soundness

- Quality of forecasting methodology
- Appropriate handling of uncertainty
- Realistic modeling assumptions

Practical Relevance

- Alignment with real ecommerce marketing workflows
- Operational usefulness for agencies and marketers
- Business interpretability of forecast

AI Integration

- Effective use of LLMs for reasoning and interpretation
- Quality of causal summaries and insights

Product Thinking

- Clarity of workflow
- Simplicity and usability
- Ability to communicate forecasts effectively

Engineering Quality

- Code quality
- Architecture clarity
- Documentation completeness
- Reliability of the prototype

Timeline

- Hackathon Start Date: **June 12, 2026**
- Submission Deadline: **July 19, 2026**
- Finalist Announcement: **July 27, 2026**
- Final Presentation: **August 5, 2026**

Resources

Date Link - [Algnition_dataset](#)

- GA4 session source/medium conversion data
- Shopify conversion data

Submissions Guidelines

- Check the attached '**Hackathon Submission Guide**' for all the details.

Conclusion

This challenge focuses on a realistic and increasingly important problem in modern digital marketing. Participants are encouraged to think beyond dashboards and reporting systems, and instead build intelligent forecasting utilities that combine statistical modeling, business reasoning, uncertainty estimation, and AI-assisted interpretation.

The strongest solutions will demonstrate technical depth, operational realism, and clear understanding of how forecasting influences real business decisions in ecommerce marketing.

All the Best!!